MAHATMA GANDHI UNIVERSITY
Fourth Semester

MODEL QUESTION PAPER - THEORY

Minor Course – MG4DSCSTA204– STATISTICS

STATISTICAL MODELLING IN DATA SCIENCE

(2024 ADMISSION)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A

Short Answer Questions

Answer any **seven** questions. Each question carries **2 marks**.

1. Mention how data science is applied in gaming. [U][1]
2. Mention any one importance of data [U][1]
3. Define discrete data and continuous data. [U][1]
4. Mention any two examples for structured data [U][1]
5. Mention any two uses of time series data. [U][1]
6. Mention the test statistic for testing the given value of a population mean in large sample case, when population SD is known. [K][2]
7. Define sampling distribution. [U][2]
8. Define alternative hypothesis with an example[K][2]
9. What is the importance of linear regression? [A][4]
10. Mention any two advantages of logistic regression [U][4] (7x2=14 marks)

Part B

Short Essay Questions

Answer any **four** questions. Each question carries **6 marks**.

11. How does decision tree work?. [A][4]
12. Explain the characteristics of transactional data. [U][1]
13. Explain any three types of biological data. [K][1]
14. A company claims its light bulb's mean life is more than 1000 hours. A sample of 22 bulbs has a mean life of 1083 hours with a standard deviation of 70 hours. Does the sample support the claim at α=0.05[A][2]
15. The SDs of two samples of size 12 and 15 from two normal populations are 3.7 and 3.2 respectively. Examine whether SDs of the populations are equal [An][3]
16. What are the benefits of big data?[U][1]

(4x6=24 marks)

Part C

Essay Type Questions

Answer any **one** question. Each question carries **12 marks**.

17. Explain categorical data, stating its advantages, disadvantages and examples [U][1]
18. Fit a Poisson distribution to the following frequency data and perform a goodness-of-fit test. [An][3]

x	0	1	2	3	4
f	17167	1861	124	2	1

(1x12=12 marks)